VINCENT MORIER-GENOUD

BIOMEDICAL ENGINEER

31 years old, Swiss

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/vincentmorier-genoud

Software Skills

Programming

Python | TensorFlow | MATLAB | Git | C | Google Cloud Platform | BigQuery

Mechanical engineering

Siemens NX | Abaqus | 3DSlicer

Other tools

FIJI | Excel (VBA) | Adobe Suite

Education

2025 **Master of Biomedical Engineering with a Major in Biomechanics**
GPA: 5,36/6
Thesis grade: 6/6
Institution: UNIBE, Bern
Master Thesis at: EPFL, Lausanne

2018 **Bachelor of Science in Automotive Engineering & *Orientation Management* Certificate**
BFH, Biel

2014 **Vocational Baccalaureate (Swiss Federal Maturity Certificate)**
CPNV, Yverdon-les-Bains

2013 **Vocational Education and Training in Automotive Mechatronics**
ETML, Lausanne

Languages

French	Native
English	Fluent (C1)
German	Intermediate (B2)

Profile

Originally trained as an automotive engineer, I discovered biomedical research while working in a laboratory at the intersection of assistive mobility and rehabilitation. Witnessing the tangible impact of technology on patients sparked a strong interest in medicine and human-centered research. This experience led me to pursue a Master's degree in Biomedical Engineering alongside my professional activity, with the goal of contributing to research at the interface between clinical practice and medical technologies. I am now fully motivated to invest in this field and to work at the forefront of healthcare innovation.

Work Experience

Master Thesis – Medical Data Scientist — 2025
Laboratory of Hemodynamics and Cardiovascular Technology, EPFL

Key Contributions: creation and curation of a hemodynamic dataset from intensive care unit patients; design and implementation of signal processing and machine learning methods applied to the collected data; development of comprehensive technical documentation; preparation of a manuscript for submission to a peer-reviewed journal.

Developed Expertise: analysis and processing of large-scale data in Python (NumPy, Pandas, SciPy); biomedical signal processing; design of machine learning models (scikit-learn, TensorFlow/Keras); management of anonymized clinical data in compliance with *Human Research – Data or Specimens Only Research standards (CITI Program).*

Scientific Collaborator — 2021-2024
SCI-Mobility Laboratory, BFH Biel

Research & Analysis: statistical analysis of the physical condition evolution of a tetraplegic individual; comparative study of mobility aids in Switzerland, including user tests with people with disabilities, assessment of autonomy, and analysis of the dynamic behavior of mobility devices.

Mechanical & Electrical Engineering: design of a test bench to measure wheelchair frame torsion; multiple strain gauge-based deformation analyses; performance testing of electric bicycle motors.

Project Management: coordination of a safety testing project for mobility assistance systems in collaboration with a crash test center (DTC AG); planning and fundraising for an electric wheelchair project integrating a system to compensate for forces applied to the user's torso.

Laboratory Assistant — 2018-2021
Institute for Energy and Mobility Research, BFH Biel

Automotive Engineering: analysis and documentation of data bus communications in electric vehicles; development of a proof of concept for a shared electric vehicle charging network; design of an embedded system for activity identification on agricultural machines using an accelerometer and FFT analysis.

Internet of Things (IoT): data transmission via telemetry (LoRaWAN); development of dashboards using the ThingsBoard platform; design of an automated agricultural irrigation valve system aimed at optimizing water usage.

Volunteering

Since I was 18, I have been actively engaged in associations, developing strong organizational skills in transdisciplinary settings:

Repris de Justesse Foundation — 2013-2022
Concert musician and technical support in charge of music stage set-up

Treasurer of the Swiss Dodgeball Federation — 2018-2020
In charge of managing the accounts, I also actively organized events, such as the 2020 European centre championship in Lausanne.